

PRACTICE ACTIVITY

Dashboard Diagnostic Lab

Calculate, Interpret, Decide

Duration: 30 minutes | Work: individual with feedback | Output: completed KPI diagnosis

Practice mission. Use the week 6 dashboard to calculate project-performance indicators, interpret what each result means, and select a corrective response. Complete each checkpoint before consulting the self-check page.

1. Learning Outcomes

Students will calculate schedule and cost indicators, interpret their direction and magnitude, connect quantitative and quality evidence, and select corrective action with immediate self-feedback.

2. Week 6 Dataset

Measure	Value
Budget at Completion (BAC)	USD 40,000
Planned progress	50%
Actual completed value	38%
Actual Cost (AC)	USD 24,000
Planned modules completed	4
Actual modules completed	2
Critical bugs open	12

3. Formula Reference

Indicator	Formula	Interpretation
Planned Value (PV)	$BAC \times \text{planned progress}$	Budgeted value of work planned
Earned Value (EV)	$BAC \times \text{actual completed value}$	Budgeted value of work completed
Schedule Variance (SV)	$EV - PV$	Negative means behind schedule
Cost Variance (CV)	$EV - AC$	Negative means over budget
Schedule Performance Index (SPI)	$EV \div PV$	Below 1 means inefficient schedule performance
Cost Performance Index (CPI)	$EV \div AC$	Below 1 means inefficient cost performance

4. Checkpoint A – Calculate PV and EV (5 minutes)

Show the calculation and include units.

PV = _____

EV = _____

In one sentence, explain why PV and EV are expressed as monetary values even though the inputs are percentages:

5. Checkpoint B – Calculate Variances (5 minutes)

SV = **EV** – **PV** = _____

CV = **EV** – **AC** = _____

Mark and justify the diagnosis:

- Schedule: ☐ ahead ☐ on plan ☐ behind
- Cost: ☐ under budget ☐ on plan ☐ over budget

Evidence-based explanation:

6. Checkpoint C – Calculate Performance Indices (5 minutes)

Round answers to two decimal places.

SPI = **EV** ÷ **PV** = _____

CPI = **EV** ÷ **AC** = _____

Complete the interpretations:

For each USD 1.00 of planned work by week 6, the project completed approximately USD _____ of value.

For each USD 1.00 spent, the project produced approximately USD _____ of earned value.

7. Checkpoint D – Integrate Non-Financial Evidence (5 minutes)

Quantitative indicators do not explain everything. Connect each additional signal to a project concern.

Signal	Concern indicated	Question for the PMO
2 of 4 modules completed		
12 critical bugs open		
Database integration risk		

8. Checkpoint E – Decide (5 minutes)

Rank these actions from 1 (strongest) to 4 (weakest):

- ☐ Add optional features to improve sponsor satisfaction.
- ☐ Freeze non-critical scope and focus on integration, essential modules, and critical bugs.
- ☐ Stop reporting until the team recovers lost progress.
- ☐ Remove acceptance testing to regain schedule time.

Justify the strongest action using at least three indicators:

9. Self-Check and Feedback

Consult this section only after attempting all checkpoints.

Expected calculations:

- $PV = \text{USD } 40,000 \times 0.50 = \text{USD } 20,000$
- $EV = \text{USD } 40,000 \times 0.38 = \text{USD } 15,200$
- $SV = \text{USD } 15,200 - \text{USD } 20,000 = -\text{USD } 4,800$
- $CV = \text{USD } 15,200 - \text{USD } 24,000 = -\text{USD } 8,800$
- $SPI = 15,200 \div 20,000 = 0.76$
- $CPI = 15,200 \div 24,000 = 0.63$

Expected diagnosis: The project is behind schedule and over budget. It has delivered only USD 0.76 of value for every USD 1.00 planned by this point and USD 0.63 of value for every USD 1.00 spent. Module completion, critical bugs, and integration exposure also show a serious quality and delivery risk.

Strongest action: Freeze non-critical scope and concentrate resources on integration, essential pilot modules, critical defects, and acceptance testing. Report the revised forecast and required decision transparently.

If your answer differed, identify whether the error came from the formula, arithmetic, sign, rounding, or interpretation:

10. Assessment and AI Boundary

Assessment: calculations (12), interpretation (8), integrated diagnosis (5), and corrective-action justification (5), totaling 30 points.

Complete calculations independently before using AI. If AI is permitted, use it to explain an error after comparing your work with the self-check. Never accept an unexplained result.